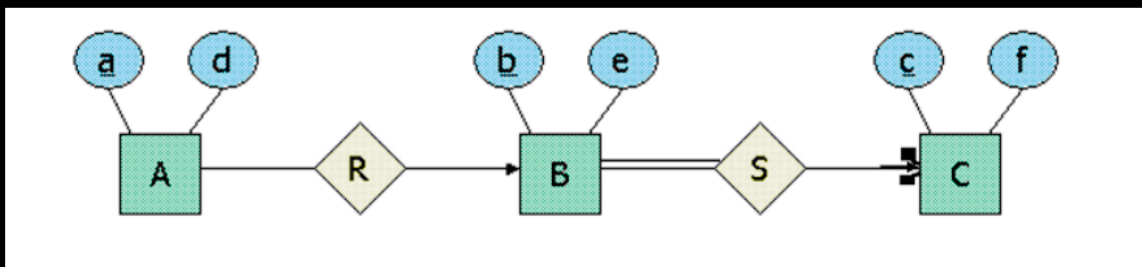


Q: A bank has a database with accounts. For each account it records the (unique) account number and the current balance. The database also has information about depositors --- their name, (unique) social-security number, and their address. Finally, the bank also stores, for each account, the depositor or depositors (in the case of joint accounts), that own the account. There is only one reasonable ER diagram for this database structure. Draw this ER diagram, and then tell which of the following appears in the ER diagram for this bank's database. 1/1

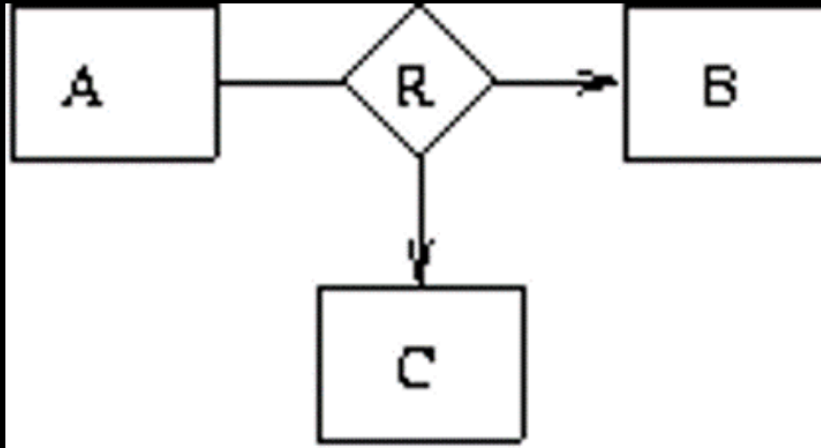
- ☐ A rectangle labeled depositor-name.
- ☒ An oval labeled address without an underline.
- ☒ A rectangle labeled Account.
- ☐ A diamond labeled Depositors.

Q: Based on the ER diagram, which of the following entity set cardinalities is valid? 1/1



- ☐ $|A| = 1; |B| = 1; |C| = 0$
- ☒ $|A| = 0; |B| = 0; |C| = 1$
- ☐ $|A| = 0; |B| = 1; |C| = 0$
- ☒ $|A| = 10; |B| = 10; |C| = 1$

Q. Let a , b , and c be the numbers of entities in entity sets A , B , and C , respectively. Let t be the number of triples in the relationship set for R . Which of the following is a possible combination of values for a , b , c , and t ? 1/1



- ☒ $a=10, b=100, c=50, t=200$
- ☐ $a=1000, b=1, c=200, t=1200$
- ☒ $a=50, b=1, c=1, t=0$
- ☐ $a=20, b=10, c=100, t=2000$

Q: Suppose we are to design a registrar's database to store information about students, courses, the courses students have taken, and the grades students have gotten in these courses. Courses have a number, a department, and a title, for example, "CS101: Introduction to computing" has department = CS, number = 101, and title = "Introduction to Computing." Numbers are assigned by departments, and different departments may use the same number. Students are represented by their (unique) student ID and their name. "Enrollments" each consist of a course, a student who took that course, and the grade the student got in the course. You should draw all the E/R diagrams that represent this database structure correctly. Then, indicate which of the following might you find in a correct E/R diagram. 1/1

- ☒ Entity set Students with attribute ID underlined and name not underlined.
- ☐ Many-one relationship Enrollments from Courses to Students, with attribute grade.
- ☐ One-one relationship Enrollments connecting Courses and Students, with attribute grade.
- ☒ Ternary relationship Enrollments connecting entity sets Courses, Students, and Grades, with an arrow to Grades.